Aalborg Universitet

AALBORG UNIVERSITY

LeichtPRO-profiles

development and validation of novel linear biocomposite structural components fabricated from pultruded natural flax fibres with plant-based resin for sustainable architectural applications

Spyridonos, Evgenia; Gerstner, Andreas; Nehmeier, Michael; Dahy, Hanaa

Published in:
Materials Research Express

DOI (link to publication from Publisher):
10.1088/2053-1591/ad6536

Creative Commons License
CC BY 4.0

Publication date:
2024

Document Version
Publisher's PDF, also known as Version of record

Link to publication from Aalborg University

Citation for published version (APA):
Spyridonos, E., Gerstner, A., Nehmeier, M., & Dahy, H. (2024). LeichtPRO-profiles: development and validation of novel linear biocomposite structural components fabricated from pultruded natural flax fibres with plant-based resin for sustainable architectural applications. *Materials Research Express*, *11*(7), Article 075308. https://doi.org/10.1088/2053-1591/ad6536

General rights
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.

- Users may download and print one copy of any publication from the public portal for the purpose of private study or research.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying the publication in the public portal -

Take down policy
If you believe that this document breaches copyright please contact us at vbn@aub.aau.dk providing details, and we will remove access to the work immediately and investigate your claim.

Downloaded from vbn.aau.dk on: 11 07, 2026

Materials Research Express

PAPER

LeichtPRO-profiles: development and validation of novel linear biocomposite structural components fabricated from pultruded natural flax fibres with plant-based resin for sustainable architectural applications

Evgenia Spyridonos[1], **Andreas Gerstner**[2], **Michael Nehmeier**[2] **and Hanaa Dahy**[1,3,4]

[1] BioMat@Stuttgart: Bio-Based Materials and Materials Cycles in Architecture, Institute of Building Structures and Structural Design (ITKE), University of Stuttgart, Keplerstraße 11, 70174 Stuttgart, Germany
[2] CG TEC Carbon und Glasfasertechnik GmbH, Gewerbepark Hügelmühle 41, 91174 Spalt, Germany
[3] BioMat@Copenhagen: Bio-Based Materials and Materials Cycles in the Building Industry Research Centre-TECH-Technical Faculty for IT & Design, Planning Department, Aalborg University, Meyersvænge 15, 2450 Copenhagen, Denmark
[4] Department of Architecture (FEDA), Faculty of Engineering, Ain Shams University, Cairo 11517, Egypt

E-mail: evgenia.spyridonos@itke.uni-stuttgart.de

Keywords: biocomposites, pultrusion, natural fibers, natural fiber pultruded profiles, natural fiber reinforced polymers (NFRP), sustainable architecture

OPEN ACCESS

RECEIVED
1 May 2024

REVISED
4 July 2024

ACCEPTED FOR PUBLICATION
18 July 2024

PUBLISHED
26 July 2024

Original content from this work may be used under the terms of the Creative Commons Attribution 4.0 licence.

Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

Abstract

Reconsidering the materials used in construction is crucial within the building industry, particularly in the context of sustainability. Recently, there has been a growing interest in exploring novel materials, with fibre-reinforced composites emerging as a prominent choice with biocomposites standing out as promising for advancing sustainability goals. This paper introduces the development of LeichtPRO-Profiles, continuous linear biocomposite profiles fabricated using the pultrusion technology. A primary focus is the application of these profiles in structural systems as load-bearing elements, emphasising the significance of understanding their mechanical properties. Specifically, an original application involves active-bending structures, necessitating a focus on the material's bending behaviour. This study discusses the methods employed in developing the pultruded biocomposite profiles which are made from natural flax fibres and an optimised matrix formulation based on a plant-based resin system. This research also outlines the optimisation of the fabrication process of these biocomposite profiles using bio-based ingredients. The results demonstrate the material's mechanical capabilities through extensive experiments and mechanical tests, revealing a compression strength of 31.2 kN and a flexural strength of 300 MPa, with a bending radius of up to 2.4 m, indicating its suitability for structural applications. Concepts of applications in several systems across different scales and contexts are also presented. The versatility and adaptability of this product make it suitable for a wide range of applications spanning various scales and thematic contexts.

1. Introduction

The building industry has a significant environmental impact on climate change. To address this, exploring various industry sectors - including methods, processes, and materials - is necessary. With the rise of sustainability awareness, eco-friendly material options are being researched and implemented, including natural and bio-based alternatives to conventional materials [1, 2]. Following the principles of the New European Bauhaus to make our built environment more sustainable, the incorporation of bio- and eco-friendly materials marks a significant step towards a greener future. Integrating such materials can not only improve the structures but also have a positive environmental impact, aligning with the objectives of the European Green Deal and the United Nations Sustainable Development Goals. This approach promotes conscious development in the industry and beyond [3–5].

© 2024 The Author(s). Published by IOP Publishing Ltd

While embracing the environmental benefits of natural building materials, it is essential to recognise their inherent limitations. Although they are entirely natural and biodegradable, their availability depends on specific geographical locations, and resource restrictions might provide difficulties. Additionally, the inconsistent cross-sections in their natural state and the length restrictions that occur due to inherent natural length require additional treatments, resulting in higher energy consumption and costs [6–8]. The growth time of organically obtained materials varies, requiring careful planning in the overall building development process. Composite materials may hold the key to overcoming these issues. Specifically, Fibre-Reinforced Composites (FRC) are materials composed of a combination of a matrix reinforced by fibres; in the case of biocomposites, at least one of these components must be obtained from renewable resources. The ability to modify the recipe and chosen ingredients to comply with specific requirements for the material itself as well as the shape and form makes bio-composite materials a viable answer to these concerns.

Various manufacturing techniques exist for producing composite and biocomposite materials and products. In recent years, extensive research has been into utilising natural fibres, most commonly flax or hemp, to create biocomposites (Natural Fibre Reinforced Composites—NFRC) for a wide range of sizes and applications. Several established technologies are employed to develop these products, such as extrusion, pultrusion, and various compression moulding techniques. Other composite manufacturing methods may involve tailored fibre placement or 3D printing, where the selected materials can be sourced from biomass. Moreover, additional research has been conducted on coreless filament winding methods to produce biocomposites of different scales using various material combinations [9, 10].

This study delves into pultrusion technology, presenting the methodologies and procedures involved in developing a new biocomposite product—a circular hollow-section profile engineered to be a load-bearing element for structural applications. It is named 'LeichtPRO-Profile', with its name referring to its lightweight nature and derives from the german word 'leicht' meaning light, and 'pro' representing profiles. This product was developed in the framework of the research project LeichtPRO (Pultruded load-bearing lightweight profiles from natural fibre composites), led by the BioMat Department at the ITKE Institute of the University of Stuttgart [11, 12]. Fabricated using pultrusion, these profiles are made from natural flax fibres and a bio-based resin, developed at a previous phase of this research. This article will delve into the material combination and optimisation of the pultrusion process at an industrial scale. The outcome is validated through mechanical testing and concepts of structural applications in various scales. Since bending structures were the focus of primary applications, particular emphasis is given to evaluating the elasticity and the bending behaviour of the produced profiles.

2. Materials and methods

2.1. Pultrusion process

Pultrusion is a technology utilised for fabricating Fibre Reinforced Polymers (FRP). It is a continuous process for producing endless profiles of consistent cross-sections. This method combines two essential components: fibres and a matrix. There are various options available for both elements. Most commonly, synthetic fibres, such as carbon or glass fibres are chosen. From the multitude of matrix options, the more predominantly used do not derive from bio-resources. Pultrusion is a method that was first established in the 1950s and it has become then a popular method in the composites industry. The geometry of the pultruded products can vary and depends on the heated mould employed in each case. Currently, the market offers multiple options of already available cross-sections for diverse applications. Carbon or glass fibre rods with square or circular geometry are among the most common pultruded products available. As pultrusion is a continuous process, the pultruded products have potentially unlimited length, primarily restricted by transportation and factory dimensions.

The pultrusion process steps are presented in figure 1. Throughout the fabrication process, fibres are continuously pulled through various components until the final product emerges at the end. Long fibre rovings are arranged on spools and are placed at the outset of the process. Depending on the type of fibre used, a fibre dryer may sometimes be integrated. The dried fibres pass through the resin bath where they become saturated by the chosen matrix system. Subsequently, the soaked fibres, guided for orientation, proceed through the heated mould where they acquire their shape. During this phase, the fibres take on their final form, and the resin undergoes curing, followed by a cooling stage. At the end of the set-up, the caterpillar take-off unit pulls the finished cured pultruded profile which is finally cut by a travelling saw to the desired length. This process operates continuously, resulting in profiles of unlimited lengths [13, 14].

The main advantages of pultruded products are their high strength-to-weight ratio and lightweight nature. A key attribute of the pultrusion process and pultruded products lies in the unrestricted length of the final product, which offers benefits for various applications, though transportation constraints persist. While diverse shapes of profile cross-sections are available, conceptually more freeform shapes can be generated, enabling customisation

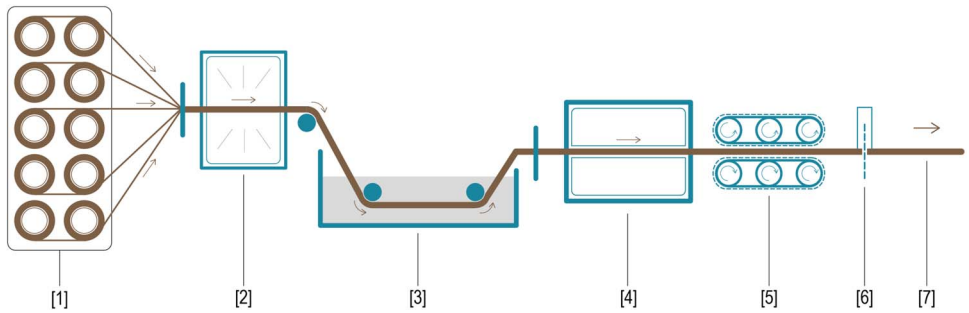


Figure 1. Pultrusion fabrication process set-up: (1) fibre roving spools, (2) fibre dryer (optional), (3) resin bath, (4) heated mould, (5) caterpillar take-off unit, (6) travelling saw, (7) cured pultruded profile, © BioMat. [23]

of desired products. However, this involves the production of new heated moulds, incurring high costs, making it impractical for market purposes to customise any desired geometry [13]. The most common shapes available are simple round and square ones. As to other composite fabrication methods, pultrusion offers the potential for modifying ingredients and components to meet specific requirements and optimise the recipe. Sustainability in composite products depends largely on material choices. Biocomposite pultruded products can offer sustainable solutions for many applications. Despite initial experimentation with natural fibres in pultrusion, there is still no successful integration of natural fibre pultruded profiles into the market [15, 16]. Thus, the scope of this project is the development of biocomposite pultruded profiles specifically fabricated for integration into structural systems as load-bearing elements.

2.2. Material development

From the beginning of the project, the goal was to develop a biocomposite pultruded profile that can be used in various structural typologies, including bending, and compressive structures. One of our main priorities was to develop active-bending structures—structures where the final geometry is achieved by bending straight or planar elements—which became one of the priorities in determining the desired material properties. The original plan involved developing two recipes for pultrusion, resulting in two distinct products: a bendable profile for active-bending systems and a stiff profile for applications under compressive loads. Nevertheless, recognising the potential in customising composite material properties for specific applications, a new challenge emerged, creating a single formulation that offers both bending and compressive strength, suitable for multiple structural uses. Despite the challenges, material development started, and the resulting profiles were tested to achieve the desired properties.

2.2.1. Definition of cross-section

The determination of the cross-section for the LeichtPRO-profiles was a thoughtful process, considering future perspectives and applications. The chosen geometry for the profiles is a circular hollow structure with a diameter of 25 mm and a wall thickness of 4 mm (figure 2). Given that pultrusion is a continuous production process, the actual length of the profiles presents no limitation. To apply the profiles in various structural systems, a round geometry proves beneficial for applications, attachments, and connection details. In particular, for gridshells or systems where element intersection or overlap is required, the round geometry minimises the connection area to a single point. This helps to allow and manage torsion in the elements and prevents surface damage that could occur with sharp corners in rectangular cross-sections. This geometry and size decision was also informed by analysing available products in the market, considering their details and connection possibilities, to ensure versatility across a broad spectrum of applications. The 25 mm diameter size was chosen because it can achieve smaller bending radii while remaining easy to handle, a characteristic that is crucial for application in bending structures. The wall thickness was determined based on the material performance and also considering minimising waste during production.

The geometry of pultruded profiles is directly linked to the available options of heated moulds. Multiple options already exist for developing market-standard products, typically utilising carbon or glass fibres. However, for this research, which focuses on utilising natural fibres, a new tool had to be developed to accommodate the specified geometry. For future scenarios reflecting different geometries in rectangular shapes or different sizes, a customised tool will be necessary. The tool is highly costly, a factor that must be considered in advance. The design of the new heated moulding tool required additional adjustments which were carried out directly after determining the profile geometry. To create a hollow geometry, the moulding tool required a core

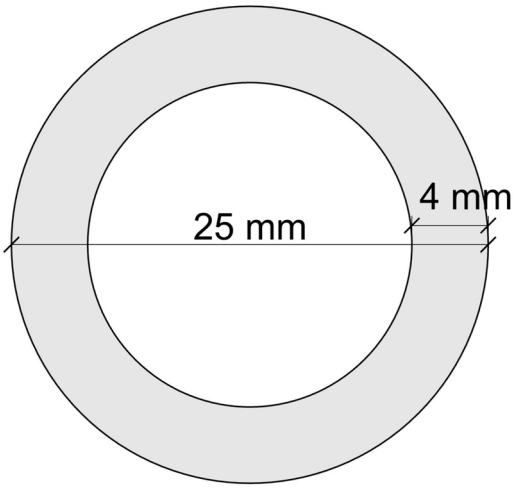


Figure 2. Profile cross-section dimension.



Figure 3. Heated moulding tool for pultrusion of round ∅25 mm profiles, © CGTEC.

suspension that needed to be centrally adjusted inside the mould. The oven also needed modifications to accommodate the hanging element. Several tests were conducted to ensure the core's central alignment throughout the entire manufacturing process, which remains challenging. Subsequently, the tool was procured, and the oven was modified accordingly, enabling the production of the profiles (figure 3).

2.2.2. Natural fibres as reinforcement

Natural fibres, particularly plant fibres, present multiple benefits when used as a reinforcement for composites. They are naturally biodegradable, have low environmental impact and recyclable raw materials with global availability at low cost [17, 18]. In this project, flax and hemp fibres are explored in the pultrusion manufacturing process to develop biocomposites. The initial stages of this research focused on conducting pultrusion trials on a laboratory scale where 6 mm round hemp-based biocomposite profiles were produced. These trials provided a strong foundation for the transition of laboratory-based pultrusion to industrial production [12].

Fibres are materials composed of natural or artificial strands woven together to form a network. Yarns, which can be continuous, short, or staple fibre strands, are fine cords made by twisting fibres together. A key factor is the linear density or yarn count, measured in tex (the weight in grams per 1000 meters of yarn). Additionally, when twisting yarns, the number of full twists per meter (tpm) is an important parameter [19, 20]. In this research, two variants with hemp fibres have been produced with a fineness of 1000 tex but different yarn twists, 37 tpm and 53 tpm. Overall, both variants are highly sensitive to tension and require careful rewinding.



Figure 4. Fibre drying oven integrated into the pultrusion line, © CGTEC.

The nature of natural fibres makes them less strong in tension compared to synthetic ones like glass fibres. To create uniform yarns with minimum breakage, a lower speed can be applied while twisting, and the final twisted yarn can achieve the desired result. The variant with 37 tpm presents a more uneven visual appearance when unwound and exhibits noticeably more protruding fibres compared to the higher-twisted variant. Occasionally, during rewinding, the 37 tpm roving would tear off. However, both variants underwent pultrusion smoothly in the laboratory tests. With the 53 tpm variant, a fibre volume content of just under 45% was achieved. For the 6 mm round bar profile (lab-scale trials), 19 hemp fibre rovings were drawn in for this purpose.

Nevertheless, to manufacture larger profiles on an industrial scale, such as those with a diameter of 25 mm, a significantly larger cross-sectional area would need to be covered with fibre reinforcement. With a fibre volume content of approximately 45%, 178 fibre rovings per 1000 tex would be required, representing a substantial increase in production costs. To mitigate this, it was advisable to utilise fibre rovings with a higher yarn count. With a yarn count of 3000 tex, only around 60 fibre rovings would be necessary, while with 4000 tex, it would be only 45 rovings. Due to market availability and time constraints, hemp and flax fibre were considered as options for the industrial-scale pultrusion process. Initially, hemp fibre of 3000 tex and flax fibres of 2400 tex and 1000 tex were used.

Various types of natural fibres underwent small-scale production tests. However, due to market availability and time constraints, flax fibres were ultimately chosen for the final production. The LeichtPRO-Profiles were ultimately manufactured using commercially available flat and untwisted flax fibre rovings from the Depestele Group. These fibres possess a linear density of 1000 tex and a thickness ranging between 0.3 mm and 0.5 mm. Their strength at break exceeds 25N, with an average mean strength of 40N. The density of the flax rovings is reported at 1.45 g/cm^3. Another crucial parameter is the moisture level, which is measured at 8%.

Natural fibres have naturally high moisture levels which can affect the mechanical properties of the resulting Natural Fibre Reinforced Polymers (NFRP). Studies have indicated compressive and tensile strength decreasing by 15% and elasticity modulus up to 20% of NFRP. Tensile strength on natural fibres is a very important parameter for pultrusion since the fibres are continuously pulled throughout the manufacturing process [21, 22]. Without pre-drying the natural fibres to reduce the residual moisture, successful resin absorption and good curing with excellent properties in the bond cannot be guaranteed. Therefore, for best results, a dryer should be integrated into the system before the impregnation (as indicated in figure 1, step 2). The temperature of the dryer should not be below 165 °C with a dwell time of at least 2.5 min. The laboratory tests conducted earlier were carried out with a drying temperature of 175 °C and a pulling speed of just under 20 cm/min, which corresponds to a dwell time of 4.5 min. However, higher drying temperatures could lead to damage to the fibres. In this case, it is crucial to conduct preliminary tests to assess any potential adverse effects. Thus, a fibre-drying oven was integrated into the industrial-scale pultrusion set-up (figure 4).

2.2.3. Matrix formulation

An integral part of this research in earlier stages was the development of a new matrix formulation, which was made specifically for the pultrusion setup, intended to maximise bio-content. This formulation was developed with a plant-based resin component of a modified Prepolymer made of Triglycerides and Polycarboxylic acids (PTP). Additional components were tested in various ratios including a short resin crosslinker, a hardener, an

Table 1. Matrix formulation.

Component	Manufacturer/Supplier	Function	Percentage [%]
PTP Resin	B.A.M. GmbH	Epoxy resin	34.6
Short Resin	B.A.M. GmbH	Crosslinker	8.6
Hardener	B.A.M. GmbH	Anhydride	33.2
Accelerator	B.A.M. GmbH	Phosphonate	4.5
Additive	B.A.M. GmbH	Stabilizer	0.7
Chemlease IC-25	ChemTrend	Internal release agent	1.1
Martinal OL-104 C	Martinswerk GmbH	Mineral filler	17.3

*The matrix formulation was developed as an integral part of the LeichtPRO research project by the German Institute of Textile and Fiber Research (DITF), in collaboration with B.A.M. GmbH [23].

accelerator and the Aluminum Trihydrate additive (ATH) which acts as a mineral filler and flame retardant. While different ratios were previously tested, in the finally optimised recipe, ATH was added at a filling level of 50%. The proportion of the components has been optimised to fit the pultrusion set-up and required outcomes through further tests in laboratory-scale pultrusion trials. The details on the matrix formulation were elaborated in an earlier phase of this work that illustrates the optimisation processes for the development of natural fibre pultruded profiles for architectural and structural applications. The details of the developed matrix formulation and its ingredients are outlined in table 1 [23].

The developed formulation comprises several individual components compared to conventional resin systems. The hardener and additives are segregated into multiple components that need to be measured and mixed in a short time. Consequently, the process is more time-consuming in industrial manufacturing technology, with higher quantities being present. However, once the matrix is mixed, its behaviour remains consistent throughout the test. This is a crucial factor to be considered in the industrial production of the profiles. The laboratory tests conducted previously provided temperature and speed values that aligned with the initial preliminary tests. Specifically, the natural fibres were dried at 165 °C for approximately 3 min and the pulling speed used was 34 cm/min with a dwell time of 2.6 min. These parameters could be successfully executed on an industrial scale with only a few additional adjustments.

2.3. Industrial pultrusion production line set-up

The pultrusion production line setup involves four main stages: production planning and scheduling, material sourcing and preparation, the actual manufacturing process, and quality control and testing of the finished product. The process begins with production planning and scheduling, where all necessary materials for production are calculated, a detailed timeline is created, and the energy consumption is calculated. After determining the required quantities, material sourcing and preparation can take place. This involves preparing the necessary number of natural fibre roving spools before inserting them into the pultrusion set-up. The ingredients for the matrix formulation are mixed according to the recipe, and then the matrix is inserted into the resin bath. Before starting the manufacturing process, the heated mould with the chosen geometry must be added to the setup, along with a fibre dryer if natural fibres are used. The appropriate speed and temperature settings must be implemented, and the required material cutting lengths must also be predefined before starting production. For the production of the final profiles, the temperature was set at 160 °C and a pulling speed of 30 cm/min was utilised, which was considered suitable for this material combination.

The industrial pultrusion manufacturing set-up is presented in figure 5. Initially, in the fibre creel the flax roving bobbins are arranged in an orderly manner. Subsequently, individual rovings are carefully guided through designated channels leading them towards the oven. During the passage through the fibre drying oven, set at 165 °C with a dwell time of 2.5 min, the residual moisture is removed from the fibre to achieve a functioning fibre-matrix adhesion. Following the drying process, the fibres are directed into the resin basin, where they are impregnated with the developed bio-based resin (table 1). The integration of subsequent guide plates is essential since they both efficiently strip off excess resin and redirect the arrangement of the fibres to form a round cross-section. As the fibres are approaching the inlet area of the mould, they are already evenly distributed over the intended pipe profile cross-section. Subsequently, the resin-impregnated fibres enter the heated moulding tool, while an internally heated core ensures the accurate curing of the inner diameter of the profile. After the curing process is completed and the profile is cooled to a certain extent, it transitions into the caterpillar haul-off, which drives the entire production line. The final step involves cutting the cured profile to the desired length by the downstream fully automatic sawing unit. This comprehensive production process ensures the creation of high-quality and precisely crafted pultruded profiles.



Figure 5. Pultrusion process on an industrial scale with integrated fibre drying system: (1) Fibre creel with 1000tex flax rovings. (2) Fibres being fed into the fibre dryer. (3) Dry fibres pass through the resin basin. (4) Guide plate for pre-positioning the fibres and wiping off excess resin. (5) Guide plate orienting the fibres evenly around the inner core. (6) Fibres entering the heated forming tool. (7) Finished cured profile. (8) LeichtPRO-Profile ∅25 mm © CGTEC.

After the production process, thorough quality control and testing must be conducted to inspect the outcomes and ensure product quality. This involves setting up tests to guarantee the quality and performance of the developed products, as well as determining transportation options and logistics for the final products. In this project, the pultrusion production was carefully planned with the developed formulation and material combination to create profiles for use in upcoming structural demonstrations. A comprehensive testing phase was conducted to validate performance based on intended applications, which will be further detailed in the following chapter.

3. Results and discussion

Upon completing an initial series of pultrusion production, the developed profiles underwent comprehensive mechanical testing to evaluate their performance. After confirming the material properties, the industrial pultrusion manufacturing process commenced, targeting applications in various structural and architectural contexts.

3.1. Performance of pultruded profiles

To assess the suitability of the pultruded profiles for structural applications, a comprehensive series of mechanical tests were conducted. These included compression and bending tests, along with a customised large-scale bending and curvature test. Additionally, a preliminary microscopic examination and a water immersion test were performed to gain a deeper understanding of the material's structure. Additionally, comparisons with bamboo, as a natural equivalent, were also included. The results of these tests are detailed in the following section.



Figure 6. Pultruded samples ∅6 mm with: (1) epoxy resin, hemp yarn, without a dryer, (2) epoxy resin, hemp rovings, without a dryer, (3) bio-resin, hemp rovings, with dryer, © CGTEC.

3.1.1. Evaluation of material combinations

To evaluate the material combinations, three series of small-scale pultrusion experiments were conducted. A limited number of pultruded profiles, with a diameter of 6 mm, were produced as samples to evaluate various combinations. As a new resin system was also developed as part of this research project, a comparison with resin systems existing in the market was essential. The tests utilised a standard commercially available epoxy resin from Olin company, specifically the Litestone 3100E resin and Litestone 3102H hardener. Additionally, the release agent Mold Wiz INT-1890M from company Pultex was used. Two series of samples were made with the standard epoxy resin, one with hemp yarn and the other with hemp rovings. In both series, the fibres were inserted into the pultrusion production line without the use of a fibre dryer. In the third series, the developed matrix formulation (table 1) was utilised in conjunction with hemp rovings that, in this case, had been dried using the dryer within the pultrusion set-up. The pultrusion parameters were set to a temperature of 130 °C and a 25 m/min pulling speed.

During the production of these samples, the suitability for pultrusion of the material combinations was assessed along with the visual appearance of the resulting profiles (figure 6). Without further evaluation of the mechanical performance of the produced samples, the focus was primarily on comparing the pultrusion process itself and the uniformity of the pultrudates. The smooth production process and the resulting samples demonstrated that the current combination of natural fibres with bio-resin can achieve a performance similar to that of the standard material combination. It is also evident that the fibre dryer plays a crucial role in pultrusion using natural fibres.

After completing the production and observation of the small-scale pultruded samples, a final test was conducted on larger profiles. This test aimed to validate the chosen material while also evaluating the performance of the new heated mould in creating profiles with a 25 mm diameter. The selected flax fibre rovings of 1000 tex were used for these samples. Two ∅25 mm profiles, each three meters long, were produced to compare the two resin systems (figure 7). One profile was produced with the same epoxy resin as in the previous test, while the other utilised the project-developed bio-resin. The pultrusion parameters were set with a temperature of 160 °C and a pulling speed of 30 m/min.

Similar to the previous test, the focus was on comparing the pultrusion process with the chosen materials in terms of production efficiency, suitability for pultrusion, and the smoothness of the final product. The produced profiles with the two different resins were first analysed for the appearance of the surface of the material. A slight visual difference was observed between the two resin systems, but this only appeared on the surface and did not affect the material's performance. As with the previous small-scale tests, the emphasis was on understanding the process and handling of the natural fibres in combination with the selected resin systems. Maintaining the central alignment of the suspended core of the moulding tool during the process proved challenging and requires optimisation, particularly when producing larger quantities of profile lengths. Drying the natural fibres before impregnation appeared crucial for production with both resin types. A temperature of 160 °C proved adequate for this purpose. A pulling speed of 30 cm/min was found suitable for this material combination and it will be used for the final production. The final decision for temperature for the final production was set across three zones at 130 °C, 160 °C, and 190 °C, respectively.

After evaluating the systematic pultrusion line, the necessary conditions have now been created to commence profile production. After all the necessary preliminary tests and preparatory work have been fulfilled, the industrial production of the profiles begins, marking the development of the manufacturing technology as complete. Further production of LeichtPRO-Profiles with the selected material combination will proceed. These developed profiles will be subject to further detailed mechanical testing in bending and compression as well as



Figure 7. Samples of ø25 mm pultruded profiles using epoxy resin (left) and the new bio-based resin (right), © BioMat.

Table 2. Compression mechanical test results.

Specimen [n = 8]	Maximum compression load (kN)	Compressive strength σ_M (MPa)	Compression modulus E_c (GPa)	Strain at compr. Strength ε_M (mm)
Sample 1	31.4	119	9.6	1.5
Sample 2	31.8	120.6	9.3	1.5
Sample 3	31.8	120.3	6.6	1.5
Sample 4	31.2	118.2	10.3	1.5
Sample 5	32.5	123.3	9.6	1.5
Sample 6	30.8	116.8	6.7	1.7
Sample 7	29.5	111.8	6.5	1.7
Sample 8	30.4	115.1	6.7	1.7
Average	**31.2**	**118.1**	**8.2**	**1.7**
Standard Deviation	0.9	3.4	1.9	0.2
Coefficient of variation	2.9%	2.9%	24.9%	12.7%

material analysis and observation. To facilitate bending tests and better understand the material's behaviour for various large-scale structural applications, longer profiles were also produced.

3.1.2. Compression mechanical tests

To evaluate the performance of the developed profiles a series of mechanical tests was carried out. Compression tests were performed in accordance with ASTM D695 [24, 25], the standard test method for compressive properties of rigid plastics [26]. The Zwick/Roell testing machine Z100 was employed and was operated at a testing speed of 1.2 mm/min with a pre-load of 0.1 MPa. A set of eight specimens of 25.4 mm in length was prepared. The main characteristics determined under this test were the maximum compression load, the compression modulus (E_c), compressive strength (σ_M) and compressive strain at compressive strength (ε_M) [27]. The results of compression tests conducted on the biocomposite profiles revealed the presence of vertical cracks, which were observed to align with the length of the fibres within the composite material. Notably, these cracks did not appear to significantly impact the overall performance of the samples. The outcome of the tests exhibited only slight variations, which were within the expected range. Further details of the test outcomes are provided in table 2. Notably, the compression strength is 118 MPa with a coefficient of variation of 3.4%. The average compression elastic modulus stood at 7.8 GPa. For context, the compressive strength of bamboo typically falls between 40 and 80 MPa, with a compressive modulus ranging from 11 to 14 GPa. However, it's

Table 3. Bending mechanical test results.

Specimen [n = 8]	Flexural strength σ_{fM} (MPa)	Bending modulus E_f (GPa)	Deformation ε_M (mm)
Sample 1	380.86	38.63	2.9
Sample 2	223.81	26.11	3.8
Sample 3	257.93	30.70	2.0
Sample 4	405.31	38.65	2.6
Sample 5	402.71	39.76	2.8
Sample 6	215.90	22.58	3.1
Sample 7	278.84	25.99	3.5
Sample 8	220.71	21.66	3.1
Average	**298.26**	**30.51**	**3.0**
Standard Deviation	84.12	7.55	0.6
Coefficient of variation	28.20%	24.74%	19.0%

important to consider factors such as species, age, and moisture content of bamboo, as these significantly influence its strength when compared with natural materials [7, 28].

3.1.3. Bending mechanical tests

3-point bending tests were performed according to EN 13706–2 [29], the standard associated with reinforced plastic composites with specifications for pultruded profiles and DIN EN ISO 14125 [30] which is about fibre-reinforced plastic composites with determination of flexural properties [26, 31]. The same testing device, the Zwick/Roell Z100 was utilised, this time set in a testing speed of 2 mm/min and a pre-load of 0.1 MPa. Eight samples were prepared for this test. To adhere to the dimensional requirements outlined in the standards for mechanical testing, the profile was longitudinally split from its tubular form into quarters. Consequently, the resulting samples had a width of approximately 15 mm, as stipulated by the standard, and a thickness of 4 mm, meeting the profile specifications. Each sample measured 80 mm in length and was positioned for testing within the 3-point bending setup with a span of 64 mm. The maximum force (σ_{fm}), Young's modulus (E_f), and deformation (ε_M) at maximum force were extracted as results. The detailed test outcomes are presented in table 3.

Notably, the bending modulus averaged 30.51 GPa with a coefficient of variation of 24.74%. The average flexural strength reached 298 MPa with a corresponding deformation of 3 mm. For comparison, the flexural strength of bamboo varies significantly, typically ranging from 50 to 150 MPa, varying between the species and age of the sample. While the full round bamboo poles are normally rigi, bamboo splits are known for their high flexibility and are frequently utilised in bending structures. Furthermore, research on bamboo fibres and fibre bundles has shown promising results for various applications, with outcomes highly dependent on the treatments used [28, 32].

A variation in flexural strength was observed among the tested samples, ranging from 215 to 405 MPa. While minor differences are typically expected during mechanical testing, this test revealed a more significant variation, particularly in samples 4 and 5. Similar variations were also noted in the bending modulus. This was reflected in the deformation behaviour of the tested samples, with those exhibiting higher strength showing smaller deformations. Specifically, samples 4 and 5 bent less during testing, correlating with their higher strength and reduced deformation. The exact cause of these differences remains uncertain, but potential factors include variations in the material's structure or minor damage to the tested samples. To ensure reliable results when testing a new material, although norms typically recommend using 3–5 specimens, we tested a larger number of samples, specifically 8 specimens, and used the overall average as the test outcome. However, further testing will be required before these profiles can be established as a new market product.

3.1.4. Large-scale bending test set-up

The initial application of the newly developed profiles focused on active-bending structures, necessitating an in-depth study of the material's bending capabilities. In these structures, initially straight elements are elastically bent until achieving the final shape. Understanding the material's behaviour during and after bending, as well as the formation of curvature, was crucial. Consequently, a large-scale bending test set-up was designed and built at the University of Stuttgart. This custom-made device was designed to accommodate the profiles with a diameter of 25 mm and a maximum length of 12 metres. The aim of this test is to explore the bending curvature of the

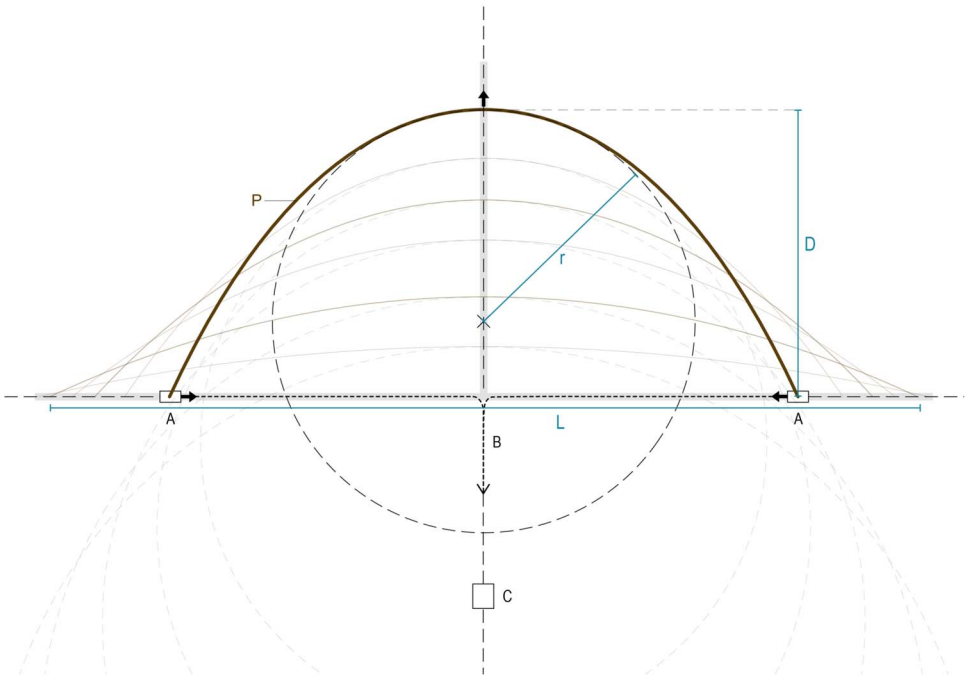


Figure 8. Bending test set-up facilitating bending and curvature tests for pultruded profiles (P) of different lengths (L) to calculate their final radius (r) and deformation (D), © BioMat.

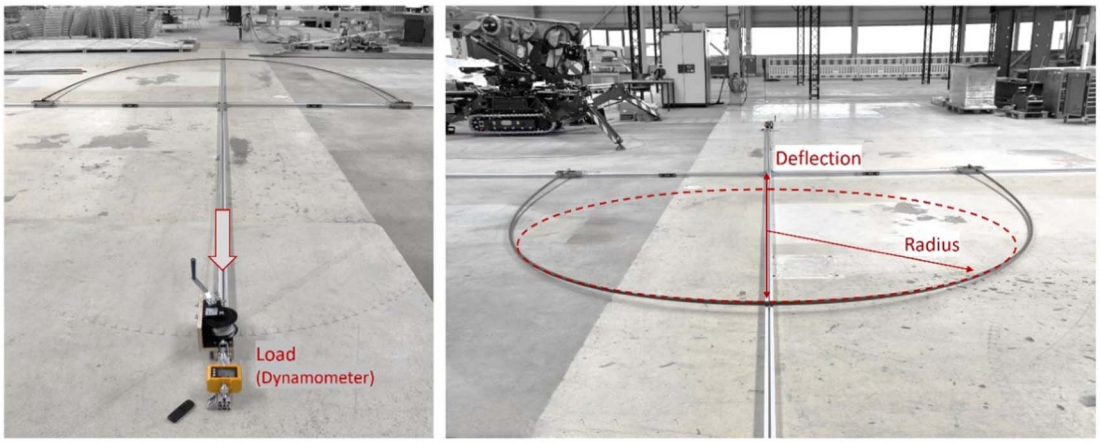


Figure 9. Large-scale bending test set-up, © BioMat.

developed profiles. As presented in figures 8 and 9, the bending process takes place in a horizontal configuration. The profile is first placed horizontally and connected at both ends (positions A). A winch (position C) is attached and connected with cables (B), pulling the two ends of the profiles towards the centre inducing the desired bend. The winch is also connected to a dynamometer (position C) to measure the resulting load. The expected outcomes of this test are the achieved radius (r), the deformation (D) and also the exerted load. Special emphasis is placed on ensuring uniform curvature through empirical observation. A ceiling-mounted camera diligently captures the bending process, providing visual documentation for comparative analysis and comprehensive cataloguing of results (figure 10).

The profiles demonstrated sufficient flexibility. After a certain length, of approximately 2 m, they could bend uniformly. The setup was designed to test the single curvature of the elements. As presented in table 4, profiles of different lengths, ranging from 7.9 m to 12.6 m were tested and catalogued. These lengths were chosen based on geometrical studies on the active-bending structure aiming to build at the next stage of this work. The resulting radii were approximately 2.4 m. Given the uniformity of the material, similar resulting radii are expected on all tested samples. Our test results indicated some variation, with radii ranging from 2.4 metres to 2.6 metres. The

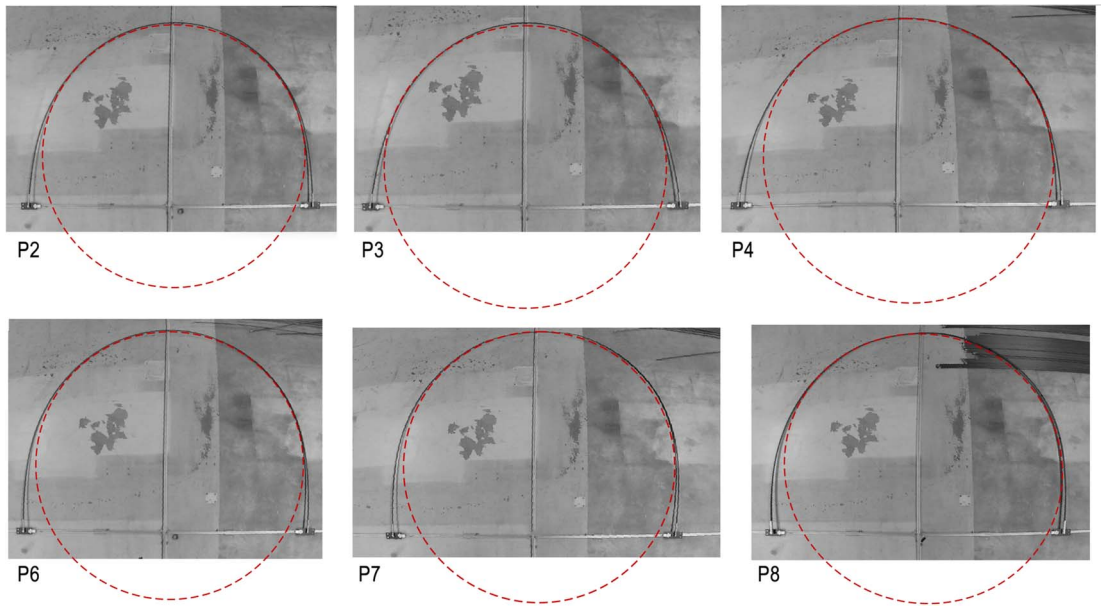


Figure 10. Bending and curvature test of profiles presented in table 4: P2 = 8.50 m, P3 = 9.75 m, P4 = 10.75 m, P6 = 12.10 m, P7 = 12.60 m, P8 = 12.60 m (double profile), © BioMat.

Table 4. Bending and curvature test results.

Profile	P1	P2	P3	P4	P5	P6	P7	P8
Length (m)	7.90	8.50	9.75	10.75	11.7	12.10	12.60	12.60
Deformation (m)	3.20	2.80	3.65	4.10	4.35	4.70	4.70	4.70
Radius (m)	2.50	2.45	2.45	2.40	2.60	2.50	2.50	2.50
Load (kg)	9.8	18	30	19	11	19.5	31	63

capture process through a camera, influenced also by perspective, can introduce inaccuracies. In future work, professional measurement tools must be utilised. At this stage, the approximation of 2.4 metres as the smallest radius can be used for the upcoming work. The deformation was derived based on the profile lengths. A set of two profiles was also included in the bending test setup, linked to the first large-scale demonstrator that integrated a double-profile perimetral contour. The double profile is the P8 depicted in figure 10. The bending was smooth without any issues when the number of profiles increased.

An additional observation after the end of the test was that the profiles did not revert to their original straight form. A slight curvature remained on them when placed freely. However, they could return to a straight form under force, which is important for transportation or packaging considerations. This characteristic was expected for a biocomposite from natural fibres because of their lignocellulosic properties, a characteristic also evident in natural materials such as timber. In contrast, bamboo returns to its original shape after bending force is applied to it as it can withstand significant bending forces [7, 33].

3.1.5. Microscopic material examination

To further understand the structural complexities of the produced biocomposite material, comparisons on a microscale were made with the naturally occurring analogue, bamboo. Bamboo's fibres align longitudinally within the material, with varying wall thickness depending on the point of the length. While considering the inherent disparities, this comparison is still relevant in the research context. To facilitate this analysis, two samples of bamboo and natural fibre pultruded profiles were examined under a microscope, aiming to gain deeper insights into their microstructure. The microscope employed for this purpose is a stereo zoom microscope, the Di-Li 914 model by Distelkamp-Electronic, with a zoom factor of 10:1 and a standard magnification range of x8 - x80.

Figure 11 presents detailed close-ups of both materials, focusing on their outer and inner surfaces, as well as their longitudinal and transverse cross-sections. In this microscopic examination, the distinct differences in microstructure between bamboo and natural fibre pultruded profiles are evident. While bamboo showcases a layered growth pattern bonded by cellulose, reminiscent of its natural origin, the pultruded profiles exhibit a

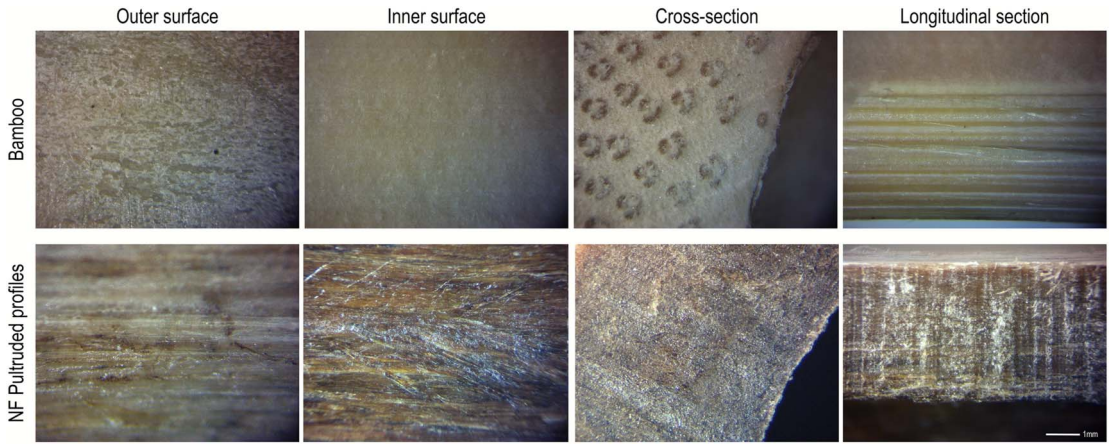


Figure 11. Comparison of fibre structure between bamboo and natural fibre pultruded profiles under microscope, © BioMat.

more homogeneous structure attributed to the manufacturing process. It is clear that bamboo, as a natural growing product, has a thicker outer surface where fibres remain concealed, while the inner surface has a smooth texture, both exhibiting a shell-like composition. The structure of bamboo's cross-section demonstrates the layered growth pattern, which is bonded by cellulose. In contrast, the pultruded profiles display uniform appearances across both outer and inner surfaces, indicative of the production process. Notably, there is no shell-like structure, which is important when considering material exposure to weather conditions. Traces of the fabrication process or remains of excess resin may also be present and should be subject to future improvements. This contrast highlights the differences between natural and synthetic materials in terms of structural complexity, mechanical performance, and environmental impact.

3.1.6. Preliminary weathering test

The developed profiles are intended for structural applications on various systems. After the first production, there was an effort to comprehend the product's response to weathering conditions, particularly when exposed to water. In a preliminary test, a number of pultruded profiles of 15 cm in length were immersed in water for a four-month period to assess water absorption, appearance changes, colour variations, and weight fluctuations. The first batch comprised of profiles precisely cut with a machine to ensure no scratches or damage. Submerged for the duration, these profiles exhibited minimal water absorption, maintaining a relatively constant weight with a marginal increase of 2%. While the surface colour shifted to a light grey shade, no significant damage was observed. Conversely, the second set comprised samples intentionally subjected to scratches and unclean cuts. Within just one month, these profiles absorbed substantial amounts of water, experiencing a weight increase of up to 60%. Surface damage was pronounced, with a noticeable shift in colour to orange.

Figure 12 presents a microscopic inspection of three profiles: a profile that remained unexposed to water, serving as a reference, the samples that displayed minimal water absorption, and the damaged profiles. At this microscale level, evident structural changes, as well as alterations in colour, are observable. The second batch of profiles shows colour degradation and minimal damage. On the contrary, the third batch exhibits noticeable damage, with exposed fibres, drying, shrinkage, and pronounced orange discolouration. This preliminary test aimed to observe and gain insights into the material's behaviour, emphasising the importance of preserving the surface integrity of pultruded profiles, especially for structural applications. Crucial is the precision in cutting composite materials, sharp cutting is necessary to avoid fibre exposure in the profile section. Moreover, safeguarding the primary structural elements from water exposure, whether rain or snow, is critical. Further scientific examinations are planned, utilising weathering machines to comprehensively observe and monitor the material's performance [34].

Due to their hydrophilic nature, natural fibre composites tend to absorb additional moisture when in contact with moisture. The orange discolouration, evident in the test, may stem from bacterial or fungal growth resulting from prolonged water exposure [35]. While the matrix shields reinforcing continuous fibres from environmental impacts, the cut edges of continuous profiles during pultrusion remain vulnerable, exposed to environmental factors and subject to ageing, limiting their service life. To prevent premature swelling or biological attacks in outdoor applications, biocomposite profiles could be subsequently sealed and shielded at cut edges with biobased lacquers or resins. Mechanical stresses such as tension, bending, impact, cutting, and abrasion can induce microcracks, allowing moisture penetration and subsequent local swelling, expansion of

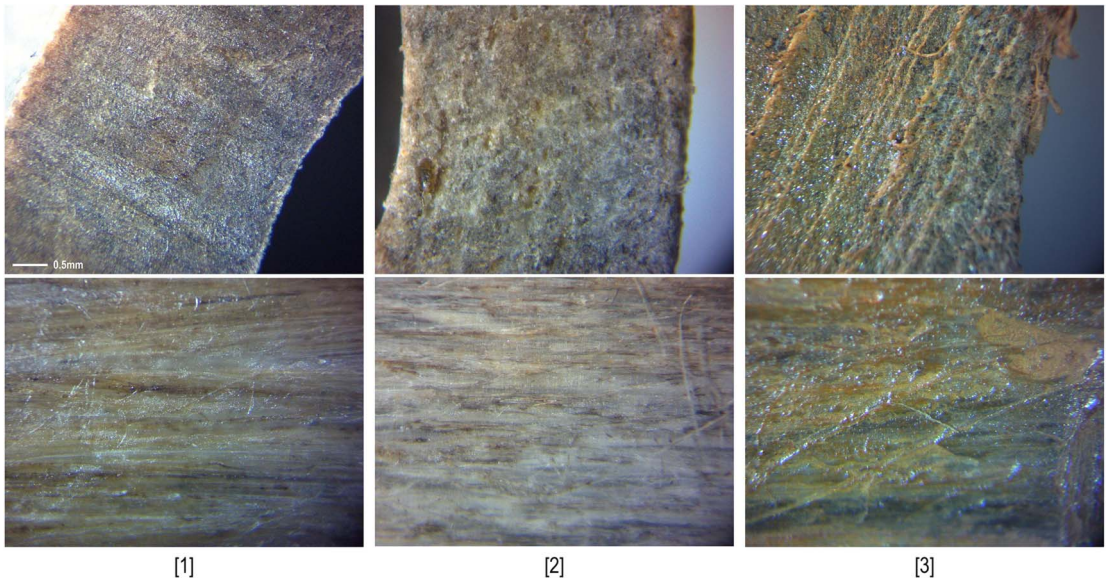


Figure 12. Comparison of cross-section and the outer surface of profile samples after immersion into water for four months: (1) unexposed to water - used as a comparison, (2) precisely cut and exposed to water - minor water absorption, (3) damaged and exposed to water, © BioMat.



Figure 13. First large-scale production of LeichtPRO-Profiles, © BioMat.

microcracks, and biological degradation, diminishing mechanical qualities and usability. One potential solution involves applying a topcoat post-production, combining various resistant properties.

3.2. Production of LeichtPRO-profiles

The pultrusion production of the LeichtPRO-Profiles was realised in several phases. Initially, a few profiles were produced as test samples and went under observation to gather feedback. After confirming the material quality after a series of tests, the final profiles were produced in multiple stages. The first large pultrusion production (figure 13) was carried out for the first realised structure. Additional material was produced which underwent a series of mechanical and geometrical tests for final validation of its performance before being applied to the structure. Since this production included profiles of greater lengths (>12 m), these profiles were validated through the large-scale bending test set-up.

Approximately 700 meters of profiles were manufactured for the first structural demonstrator, the LightPRO Shell, a ten-meter span active-bending structure which was developed at the University of Stuttgart [36]. The geometry of the structure required profiles ranging in length from 7 to 12.5 meters. Transportation constraints influenced the maximum lengths possible and thus the shape of the structure, also defining the production process. Material waste was reduced significantly by taking advantage of the continuous nature of

Table 5. LeichtPRO-profiles properties.

Product properties	
Material Name	LeichtPRO-Profile
Type of material	Natural Fibre Pultruded Profile
Category	Natural Fibre Reinforced Polymers (NFRP)
Manufacturing process	Pultrusion
Fibre	Flax fibres 1000–2400 tex
Matrix	Customised bio-based resin (table 1)
Fibre-volume ratio	≈40%
Cross-section	Circular and hollow
Dimensions	outer ∅ 25 mm, inner ∅ 17 mm
Wall thickness	4 mm
Length	unlimited
Weight	0.35 kg/m
min bending radius	≈2.4 m
Bending strength	30.5 MPa
Compression strength	118 MPa

pultrusion. To simplify the process, the profiles were divided into seven different length groups during production. The overall production was then transferred to the prefabrication location before finally being moved to the building site.

A subsequent large-scale pultrusion production generated a total of 220 meters of profiles. This iteration utilised the latest recipe, incorporating refinements derived from previous trials. A portion of these profiles was used for the construction of a 5-meter span reciprocal canopy, which was showcased at the Venice Biennial 2023, organised by the European Cultural Centre (ECC) [37, 38]. The remaining profiles, that were not used for these two structures, were employed for smaller tests and experiments in different systems to showcase and prove the capacity of the developed material.

3.3. Product characterisation

LeichtPRO-Profiles are natural fibre pultruded profiles, classified under the category of Natural Fibre Reinforced Polymers (NFRP), manufactured through the pultrusion process. These profiles utilise natural flax fibres, in 1000 tex or 2400 tex, impregnated in a customised bio-based resin formulation, resulting in a fibre-volume ratio of approximately 40%. The profiles feature a circular, hollow cross-section with an outer diameter of 25 mm, an inner diameter of 17 mm, and a wall thickness of 4 mm, weighing 0.35 kg per metre. They can be manufactured in unrestricted lengths, though current productions have been limited to 12.5 metres due to transportation constraints. The profiles have a minimum bending radius of approximately 2.4 metres. Their mechanical properties include a bending strength of 30.5 MPa and a compression strength of 118 MPa. The properties of the profiles are summarised in table 5. The data presented establish a foundational basis for further analysis and optimisation, guiding subsequent steps in the material development process. Additionally, this info serves as a basis for structural and architectural applications to follow.

3.4. Structural and architectural applications

The properties of the developed pultruded profiles demonstrate their suitability for a multitude of applications. There is a wide range of potential applications spanning various scales and thematic contexts, demonstrating the versatility and adaptability of this product. Current research is actively investigating the possible uses of these profiles across diverse contexts. The most significant example to date is the LightPRO-Shell, an active-bending structure with a covering span of 10 m, the first large-scale structural demonstrator utilising the novel profiles developed at the University of Stuttgart (figure 14(1)) [39]. This initial demonstrator highlighted the significance of understanding the material's bending behaviour during the material development and testing phases. Other prototypes, featuring various structural systems in different scales, have already been created or are in progress. These prototypes encompass architectural applications, including a facade supporting structure, tensegrity systems, deployable and reciprocal structures, as well as a dome demonstrator. In all these systems the profiles are used as load-bearing elements in various scenarios under compressive or bending loads. Various connection strategies have been thoroughly examined, including existing market-available connections, customised solutions, and experimental approaches incorporating biocomposites and natural materials. The following images, presented in figure 14, provide an overview of the latest developed demonstrators using the pultruded



Figure 14. LeichtPRO-profiles—structural applications 2021–2023 | (1) LightPRO-Shell: active-bending structure, (2) Tensegrity FlaxSeat, (3)(4) Delta |Xl: deployable system in furniture scale and facade application, (5)(6) Reciprocal canopy in two configurations, (7) Biocomposite facades support structure, (8) Buckyball Dome with biocomposite connectors, © BioMat.

profiles. The methods and techniques utilised in realising these structures will be comprehensively detailed in forthcoming scientific articles.

4. Summary

This paper details the methods used in developing the LeichtPRO-Profiles, linear biocomposite pultruded profiles. Pultrusion is a well-established fibre-reinforced composites manufacturing technology, typically employing synthetic fibres. The initial phase of this work involved creating a novel recipe aimed at reducing environmental impact for more sustainable construction. Natural fibres, specifically untwisted flax fibres, were chosen after research into various types, including hemp and flax. Additionally, a customised matrix formulation based on a plant-based resin system was developed in an earlier face of this work and is presented here. The pultrusion line was optimised to facilitate production with the selected bio-based materials. A fibre dryer was integrated into the pultrusion process to remove excessive humidity from the natural fibres, which is crucial for producing strong composites. Considering the intended applications and market availability in connector options, a round and hollow geometry with a diameter of 25 mm and a wall thickness of 4 mm was defined. To achieve this, a new heated mould with a suspended core was developed and integrated into the set-up. The two elements were incorporated into the overall pultrusion production process.

The developed profiles are intended for use as load-bearing elements in various architectural and structural applications. This required extensive testing to understand the material's mechanical properties. Initially, small-scale production of a few profiles was carried out to test different material combinations in pultrusion. Comparisons were made between materials with epoxy and bio-based resin and different types of natural fibres, particularly hemp and flax. These tests demonstrated similar pultrudability across different matrices, confirming the material's performance during production and highlighting the need for a fibre dryer and specific pultrusion parameters.

Following the initial observations, mechanical compression and bending tests were conducted to determine the profiles' properties. Their compression strength reached 31.2 kN and their flexural strength 300 MPa. A comparison with bamboo, due to its similar geometry and material nature, showed bamboo's compression strength is higher, ranging between 40 and 80 kN, while its flexural strength is significantly lower, usually not exceeding 150 MPa. Bamboo's properties, however, are influenced by factors such as species, age, and wall thickness. Given the initial application concept on active-bending structures, an in-depth study of the material's bending capabilities was necessary. Thus, an in-house large-scale bending test set-up was developed to observe bending behaviour and curvature. The profiles demonstrated significant flexibility, achieving a minimum radius of approximately 2.4 m. In shorter sections, below 2 m, the profiles exhibited good compression behaviour, suggesting potential applications in various compression structures. Preliminary microscopic tests were also conducted to compare the profiles' structure with bamboo, highlighting differences in microstructure due to the homogenous manufacturing process of the profiles compared to bamboo's layered growth pattern. Finally, a

preliminary weathering test, including water immersion and observation, provided important insights into the need to protect the composite material from moisture to prevent swelling or biological attacks.

The final production of the pultruded profiles occurred in two main phases, producing approximately 1 km of natural fibre pultruded profiles intended for testing and architectural and structural applications. The most significant application to date is a ten-metre span active-bending structure, the first large-scale demonstrator. Other completed or developing structures include a biocomposite facade supporting structure, tensegrity systems, deployable and reciprocal structures, and a dome demonstrator. Ultimately, the versatility and adaptability of this product make it suitable for a wide range of applications across various scales and contexts.

5. Conclusion

Pultruded profiles, in general, offer versatility with their ability to be developed in various diameters and cross-sections, providing a wide range of application possibilities. By customising the formulation, profiles can be optimised for specific uses, overcoming challenges typically associated with natural materials such as variable growth times and inconsistent geometries and structural properties. However, diversifying geometries necessitates the creation of new moulding tools, incurring substantial costs. The development of custom recipes is typically constrained, making it challenging for customers to define their formulations easily. In the market, standardised materials with widely accepted geometries and properties cater to common applications with defined mechanical properties. Additionally, the lengths of pultruded profiles are unlimited, and customisable to suit specific applications, thereby minimising material waste. Length constraints are primarily tied to factory size and transportation logistics. Unlike naturally grown materials, for example bamboo, which face geographic limitations, this material is unrestricted from such constraints. With pultrusion plants capable of operating worldwide, production can overcome such restrictions. Natural fibres, a key component in this material, are abundant globally and often sourced from agricultural waste.

The development of LeichtPRO-Profiles has demonstrated the feasibility of creating strong, durable biocomposite profiles. Using natural flax fibres and a plant-based resin system combined with an optimised pultrusion process, yield profiles with impressive mechanical properties, including a compression strength of 31.2 kN and a flexural strength of 300 MPa. The significant flexibility in longer lengths and good compression behaviour in shorter sections indicate their suitability for various architectural and structural applications, such as active-bending and compression structures. While it has not yet reached the market as a finalised product, the current findings represent a preliminary impression of the material's capabilities. To refine our understanding and pave the path towards a market-ready product, additional rounds of pultrusion tests are required to delve deeper into the material's behaviour under diverse conditions, allowing for the determination of specified mechanical properties that meet industry standards. This innovative work not only meets rigorous mechanical demands but also showcases the potential of eco-friendly composites in various contexts. Future perspectives involve further optimisation of the material and production processes to enhance durability and performance, with a focus on improving moisture resistance and exploring additional applications. Advanced protection methods against environmental factors will be crucial for long-term performance. Continued research will refine these materials and expand their functional capabilities, potentially leading to their broader adoption and integration into mainstream construction practices. The versatility and sustainability of the novel profiles position them as a promising alternative to traditional synthetic composites, offering significant potential for innovative, eco-friendly construction solutions and contributing significantly to the industry's sustainability goals.

Acknowledgments

The work presented in this paper is part of the results of the research project LeichtPRO: Pultruded load-bearing lightweight profiles from natural fibre composites, where all authors were involved. BioMat/ITKE at the University of Stuttgart was the project administrator and was responsible for the design development, material testing and construction of the research demonstrators. The Deutsche Institute für Textil- und Faserforschung (DITF) and Bio-Composites And More GmbH (B.A.M.) developed the bio-based resin and processes for the production of the pultruded components and conducted the laboratory scale tests. CG TEC Carbon und Glasfasertechnik GmbH was the industrial partner that handled the pultrusion manufacturing process on an industrial scale. Zenvision GmbH supported the architectural design development and Steinhuder Werkzeug developed the tools used for industrial fabrication. The authors would like to express their gratitude to all project partners for their support and valuable input throughout the project. They would also like to thank all BioMat team members, partners and students who contributed to the realisation of the research demonstrators.

Funding

This research was funded by the Fachagentur Nachwachsende Rohstoffe e V (FNR, Agency for Renewable Resources) under Bundesministeriums für Ernährung und Landwirtschaft (BMEL, Federal Ministry of Food and Agriculture) throughout the research project LeichtPRO: Pultruded load-bearing lightweight profiles from natural fibre composites (FKZ:22027018), managed by BioMat Department at ITKE, University of Stuttgart.

Data availability statement

The data that support the findings of this study are available upon reasonable request. The data that support the findings of this study are available upon reasonable request from the authors.

CRediT authorship contribution

Evgenia Spyridonos: Conceptualization, Methodology, Validation, Investigation, Writing - original draft, Writing - review & editing, Visualization, Project administration. **Andreas Gerstner**: Validation, Investigation, Writing - Review & Editing. **Michael Nehmeier**: Validation, Investigation, Writing - Review & Editing. **Hanaa Dahy**: Conceptualization, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Awards

The LeichtPRO-Profiles have received the material price (materialPREIS 2021) for their application in the first large-scale structural demonstrator, the LightPRO Shell [40]. This structure has garnered recognition through various awards including: the longlist for the Dezeen Awards 2022 [41], a nomination for the German Sustainability Award 2022 [42], a nomination for the Bundespreis Ecodesign 2022 [43], and a win at the Green Product Awards 2022 [44].

ORCID iDs

Evgenia Spyridonos https://orcid.org/0000-0002-7786-4283
Hanaa Dahy https://orcid.org/0000-0003-3428-0074

References

[1] European Commission 2020 Circular economy action plan (https://environment.ec.europa.eu/strategy/circular-economy-action-plan_en) (accessed December 15, 2023)
[2] Horn R, Dahy H, Gantner J, Speck O and Leistner P 2018 Bio-inspired sustainability assessment for building product development—concept and case study *MDPI Sustainability* **10** 130
[3] New European Bauhaus - European Union 2024 (https://new-european-bauhaus.europa.eu/about/about-initiative_en) (accessed February 6, 2024)
[4] The European Green Deal - European Commission 2021 (https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en) (accessed February 6, 2024)
[5] United Nations Environment Programme 2020 Global Status Report for Buildings and Construction - Executive Summary, UN Environment Programme (UNEP) (https://globalabc.org/resources/publications/2020-global-status-report-buildings-and-construction) (accessed January 10, 2022)
[6] Allen E and Iano J 2019 *Fundamentals of Building Construction: Materials and Methods* (Wiley)
[7] 2015 Bamboo: the Plant and its Uses *Bamboo* Liese W and Köhl M (ed) (Springer International Publishing) (https://doi.org/10.1007/978-3-319-14133-6)
[8] Shmulsky R and Jones P D 2019 *Forest Products and Wood Science: An Introduction* (Wiley)
[9] Dahy H 2019 Materials as a design tool' design philosophy applied in three innovative research pavilions out of sustainable building materials with controlled end-of-life scenarios *Buildings* **9** 64
[10] Dahy H 2023 Materials as design tool: digital fabrication strategies for sustainable architecture *Technology|Architecture + Design* **7** 153–8
[11] Fachagentur Nachwachsende Rohstoffe e. V. 2023 (https://biowerkstoffe.fnr.de/projekte/projektuebersicht/projekte-details?fkz=22027018) Pultruded load-bearing lightweight profiles made of natural fibre composites (LeichtPRO)(accessed March 25, 2024)
[12] Dahy H, Spyridonos E, Witt M U, Kaiser P, Gerstner A and Dippon K 2023 (https://fnr.de/ftp/pdf/berichte/22027018.pdf) Pultrudierte tragende leichtbauprofile aus naturfaserverbundstoffen, akronym: LeichtPRO, fachagentur nachwachsende rohstoffe (FNR)
[13] Joshi S C 2012 12 - the pultrusion process for polymer matrix composites *Manufacturing Techniques for Polymer Matrix Composites (PMCs)* (Woodhead Publishing) (https://doi.org/10.1533/9780857096258.3.381)
[14] Witt M U, Milwich M, Gresser G T, Hammer M and Knippers J 2015 Bio-pultrusion of façade-profiles from sustainable materials *Mater. Sci. Forum* **825–826** 1033–8

[15] Gengnagel C, Hernández E L and Bäumer R 2013 Natural-fibre-reinforced plastics in actively bent structures *Proceedings of the Institution of Civil Engineers - Construction Materials* **166** 365–77
[16] Minchenkov K, Vedernikov A, Safonov A and Akhatov I 2021 Thermoplastic pultrusion: a review *MDPI Polymers* **13** 180
[17] Pickering K 2008 *Properties and Performance of Natural-Fibre Composites* (Elsevier)
[18] Singh Y, Kumar J, Singh I and Rakesh P K 2022 3 - joining behavior of natural fiber reinforced polymer composites *Joining Processes for Dissimilar and Advanced Materials* (Woodhead Publishing) (https://doi.org/10.1016/B978-0-323-85399-6.00018-7)
[19] Mao N and Russell S J 2015 Chapter 13 - fibre to fabric: Nonwoven fabrics *Textiles and Fashion* (Woodhead Publishing) (https://doi.org/10.1016/B978-1-84569-931-4.00013-1)
[20] Li D (X) 2020 3 - Fundamental of fibers *Cut Protective Textiles* (*The Textile Institute Book Series*) ed D (X) Li (Woodhead Publishing) 59–110
[21] Abdollahiparsa H, Shahmirzaloo A, Blok R and Teuffel P 2023 Influence of moisture absorption on tensile and compressive properties of natural fiber-reinforced thermoplastic composites *Polymer-Plastics Technology and Materials* **62** 2138–42
[22] Ferreira S R, Mendes de Andrade R G, Koenders E, de Andrade Silva F, de Moraes Rego Fairbairn E and Toledo Filho R D 2021 Pull-out behavior and tensile response of natural fibers under different relative humidity levels *Constr. Build. Mater.* **308** 124823
[23] Spyridonos E, Witt M-U, Dippon K, Milwich M, Gresser G T and Dahy H 2024 Natural fibre pultruded profiles: illustration of optimisation processes to develop high-performance biocomposites for architectural and structural applications *Composites* C **14** 100492
[24] ASTM D695-23 2023 Standard test method for compressive properties of rigid plastics
[25] D20 Committee 2023 Test Method for Compressive Properties of Rigid Plastics, ASTM International, (https://doi.org/10.1520/D0695-1510.1520/D0695-15)
[26] Ding L, Liu X, Wang X, Huang H and Wu Z 2018 Mechanical properties of pultruded basalt fiber-reinforced polymer tube under axial tension and compression *Constr. Build. Mater.* **176** 629–37
[27] Ramamurty U 2001 Mechanical testing methods of fibers and composites *Encyclopedia of Materials: Science and Technology* 2nd edn, ed K H J Buschow (Elsevier) pp 5271–4
[28] Liese W. 1985 Anatomy and properties of bamboo *Recent Research on Bamboos* (Wiley)
[29] DIN EN 13706-2:2003-02 2002 Reinforced plastic composites - specifications for pultruded profiles - Part 2: methods of test and general requirements; German version EN 13706-2
[30] DIN EN ISO 14125 2011 Fibre-reinforced plastic composites - determination of flexural properties (ISO 14125:1998 + Cor.1:2001 + Amd.1:2011); German version EN ISO 14125:1998 + AC:2002 + A1
[31] Scărlătescu D D, Modrea A and Stanciu M D 2019 Three-point bend test to determine the mechanical behavior of the tubes used in water supply networks *Procedia Manufacturing* **32** 179–86
[32] Gao X, Zhu D, Fan S, Rahman M Z, Guo S and Chen F 2022 Structural and mechanical properties of bamboo fiber bundle and fiber/bundle reinforced composites: a review *Journal of Materials Research and Technology* **19** 1162–90
[33] Desch H E and Dinwoodie J M J M 2016 *Timber: Structure, Properties, Conversion and Use* (Bloomsbury Publishing)
[34] Das S C, La Rosa A D, Goutianos S and Grammatikos S 2023 Effect of accelerated weathering on the performance of natural fibre reinforced recyclable polymer composites and comparison with conventional composites *Composites Part C: Open Access* **12** 100378
[35] Muñoz E and García-Manrique J A 2015 Water absorption behaviour and its effect on the mechanical properties of flax fibre reinforced bioepoxy composites *Int. J. Polym. Sci.* **2015** e390275
[36] LightPRO Shell 2021 Active-bending structure from natural fiber pultruded profiles (https://vimeo.com/636947389)
[37] Spyridonos E, Reiner A and Dahy H 2024 Reciprocal lightweight structures with natural fiber biocomposite profiles through computational design and case studies validation *Proceedings of IASS 2024 International Association for Shell and Spatial Structures* Zurich Switzerland, 2024
[38] ECC Italy 2023 Venice Biennial Architecture Exhibition2023 (accessed March 21, 2024) (https://ecc-italy.eu/exhibitions/2023archbiennial)
[39] BioMat Pavilion 2021 LightPRO Shell | Institute of Building Structures and Structural Design | Uuniversity of Stuttgart (accessed January 16, 2024) (https://itke.uni-stuttgart.de/research/built-projects/biomat-pavilion-2021/)
[40] LightPRO Shell | materialPREIS 2021 Raumprobe (2021) (https://raumprobe.com/de/projekte/biomat-pavillon-2021-leichtpro-shell-934)
[41] Dezeen, LightPRO Shell | Dezeen Awards 2022 Dezeen (2022) (https://dezeen.com/awards/2022/longlists/lightpro-shell/) (accessed February 5, 2024)
[42] LightPRO Shell | German Sustainability Awards 2023 (https://nachhaltigkeitspreis.de/design/preistraeger-design/2022/nominierte/) Deutscher nachhaltigkeitspreis(accessed February 5, 2024)
[43] LightPRO Shell | German Ecodesign Award 2022 Bundespreis Ecodesign (2022) (https://bundespreis-ecodesign.de/en) (accessed February 5, 2024)
[44] LightPRO Shell | Green Product Award 2022 Green Product Award (2022) (https://gp-award.com/en/produkte/lightpro-shell) (accessed February 5, 2024)